Advanced SQL

Null Functions - Exercises

Table 1 : Employees

Q1. Show all employees with their salary. If salary Null, display 0.

```
SELECT    employee_id,
          name,
          IFNULL(salary, 0) AS salary_with_default
FROM      Employees;
```

employee_id	name	salary_with_default
1	Alice	5000
2	Bob	0
3	Charlie	7000
4	Dana	0

Q2. Show employees names with their department. If department is Null, show 'Not Assigned'.

```
SELECT    employee_id,
          name,
          IFNULL(department, 'Not Assigned') AS department_name
FROM      Employees;
```

employee_id	name	department_name
1	Alice	HR
2	Bob	IT
3	Charlie	Not Assigned
4	Dana	Finance

Table 2 : Orders

Q3. Find ~~th~~ orders with NULL customer_id using ISNULL()

```
SELECT    order_id,
          customer_id
FROM      Orders
WHERE     ISNULL (customer_id) ;
```

order_id	customer_id
103	Null

Q4. Show all orders. If delivery_date is Null, show 'Pending'.

```
SELECT    order_id,
          customer_id,
          COALESCE (delivery_date, 'Pending') AS delivery_status
FROM      Orders ;
```

order_id	customer_id	delivery_status
101	201	2024-12-01
102	202	Pending
103	Null	2024-12-03

Table3 : Students

Q5 Show all students and their grades. Replace Null with 0.

```
SELECT    student_id,
          name,
          IFNULL (grade, 0) AS final_grade
FROM      Students ;
```

student_id	name	final_grade
1	Ethan	85
2	Maya	0
3	Olivia	90

Q6. Count students who haven't been graded.

```
SELECT    COUNT(*)    AS    not_graded_count
FROM      Students
WHERE     grade IS NULL;
```

not_graded_count

1

Table 4 : Products

Q7. Show product name, price and final price after discount (assume 0 if discount is NULL).

```
SELECT    product_id,
          name,
          price - IFNULL(discount, 0) AS final_price
FROM      Products;
```

product_id	name	final_price
501	Keyboard	25
502	Mouse	10
503	Monitor	100

Table 5 : Customers

Q8. Count how many customers have no email

```
SELECT    COUNT(*)    AS    missing_email_count
FROM      Customers
WHERE     email IS NULL;
```

missing_email_count
2

Q9. Show all customers with email. If Null, display 'No Email'.

```
SELECT    customer_id,
          name,
          IFNULL(email, 'No Email') AS email_display
FROM      Customers;
```

customer_id	name	email_display
1	Linda	No Email
2	Joseph	joseph@mail.com
3	Mia	Null

Table 6 : Payments

Q10. Show payment details with method replaced by 'Unknown' if NULL

```
SELECT    payment_id,
          IFNULL(method, 'Unknown') AS method_display,
          status
FROM      Payments;
```

payment_id	method	status
301	Credit	Null
302	PayPal	Success
303	Unknown	Failed

Table 7 : Inventory

Q11. Show items and their quantity (0 if Null)

```
SELECT    item_id,
          item_name,
          IFNULL(quantity, 0) AS quantity_checked
FROM      Inventory.
```

item_id	item_name	quantity_checked
1	Pen	0
2	Notebook	150
3	Eraser	0

Table 8 : Employees_Extra

Q12. Show employee ID and the first available value among bonus or commission.

```
SELECT    emp_id,
          COALESCE (bonus, commission) AS first_available_reward
FROM      Employees_Extra ;
```

emp_id	first_available_reward
1	300
2	100
3	Null

Table 9 : Classes

Q13. Count classes that don't have a room assigned

```
SELECT    COUNT (*)  AS  no_room_count
FROM      Classes
WHERE     room IS NULL ;
```

no_room_count
2

Table 10 : Attendance

Q14. Show attendance records with status. Replace NULL with 'Not Marked'.

```
SELECT    student_id,
          date,
          IFNULL (status, 'Not Marked') AS attendance_status
FROM      Attendance ;
```

student_id	date	status
1	2025-04-01	Not Marked
2	2025-04-01	Present
3	2025-04-01	Absent

Table 11 : Bank_Accounts

Q15. Show account ID, account_type (or 'Unknown') and balance (or 0).

```
SELECT  account_id,
        IFNULL(account_type, 'Unknown') AS type_display,
        IFNULL(balance, 0) AS balance_checked
FROM    Bank_Accounts;
```

account_id	type_discovery	balance_checked
A1	Savings	0
A2	Current	5000
A3	Unknown	2000

Table 12 : Projects

Q16. Show all projects with a start date. If start_date is NULL, display 'TBD'.

```
SELECT  project_id,
        title
        IFNULL(CAST(start_date AS STRING), 'TBD') AS start_display
FROM    Projects;
```

project_id	title	start_display
1	Website Revamp	2025-10-10
2	Mobile App	TBD
3	Data Migration	TBD

Table 13 : Reviews

Q17. Display reviews showing comment (or 'No comment') and rating (or 0)

```
SELECT    review_id,
          product_id,
          IFNULL (comment, 'No comment') AS comment_display,
          IFNULL (rating, 0) AS rating_display
FROM      Reviews;
```

review_id	product_id	comment_display	rating_display
1	501	Great product	4
2	502	No comment	0
3	503	Works fine	3

Table 14 : Suppliers

Q18. Show the supplier contact number. Use COALESCE (phone, alt_phone, 'No contact').

```
SELECT    supplier_id,
          name,
          COALESCE (CAST (phone AS STRING), CAST (alt_phone AS STRING, 'No
          contact') AS contact_number
FROM      Suppliers;
```

Supplier_id	name	contact_number
1	Global Goods	123456789
2	Best Supplies	987654321
3	ValueSource	No contact

Table 15 : User_settings

Q19. Show all users and their preferences. Replace all NULLS with default:

- Theme → 'Light'
- Language → 'English'
- Timezone → 'UTC'

```
SELECT  user_id,
        IFNULL(theme, 'Light') AS theme_set,
        IFNULL(language, 'English') AS language_set,
        IFNULL(timezone, 'UTC') AS timezone_set
FROM User_Settings;
```

user_id	theme_set	language_set	timezone_set
1	Light	English	UTC
2	Dark	English	UTC +1
3	Light	English	UTC

Table 16: Maintenance

Q20. Show maintenance log with:

- issue → default to 'Unknown Issue'
- technician → default to 'Not Assigned'

```
SELECT  record_id,
        machine_id,
        IFNULL(issue, 'Unknown Issue') AS issue_log,
        IFNULL(technician, 'Not Assigned') AS technician_name
FROM Maintenance;
```

record_id	machine_id	issue_log	technician_name
1	M101	Overheating	Not Assigned
2	M102	Unknown Issue	Not Assigned
3	M103	Jammed	Alex